

Chuyue Li

lichuyue0312@gmail.com (preferred) | chuyuel2@illinois.edu

Research Interest: Agentic RL, Multimodal Generation/Understanding/Reasoning, Self-Evolve Agents

Education

University of Illinois Urbana-Champaign MS of Computer Science(MSCS)	2026.08 - 2028.06
University of California, Berkeley CS Junior <i>Exchange</i> GPA: 3.9/4.0	2024.08 - 2025.05
ShanghaiTech University CS <i>Undergrad</i> Major GPA 3.87/4.0	2022.9 - 2026.7

Internship Experience

Alibaba Qwen | Research Intern 2025.06-2025.11

Self-Evolving Agent for Video Understanding and Representation Learning
Visual Technology Department, AIGC Team *Mentor: Jingpeng Yu, Jiaming Liu*

- **First-Authored Paper Under Review:** AVA-Encoder: Towards Agent-Native Video Representation Learning.
- Served as first author; pioneered agentic video auto-encoding by representing videos as a structured Knowledge Graph. Enabled agents to encode videos into structured graphs and reconstruct them via a fixed decoder, optimizing the system through textual-gradient feedback based on reconstruction consistency.
- Designed a Hierarchical Agentic Encoder to minimize information loss. Developed a dual-loop optimization mechanism: an Outer Loop with an Anti-Forgetting Gate to learn transferable encoding policies, and an Inner Loop with an Anti-Degradation Gate for test-time KG refinement.
- Achieved SOTA reconstruction fidelity (49.0% overall, a 20.7% absolute gain over baseline). Notably, the self-learned policy outperformed human-tuned version (45.8% vs. 44.4%) while reducing token consumption by 74.3% (shot level) and 70.1% (keyframe level), enabling highly efficient, topology-aware editing.

Tencent Hunyuan | Research Intern 2025.06-2025.11

Multi-level Auto-Planning Interactive Visual Generative Agent
Multimodal Model Department, Visual Agent Team *Mentor: Rongwei Quan*

- **Accepted by CVPR 2026 Findings:** VisionCreator: A Native Visual-Generation Agentic Model with Understanding, Thinking, Planning and Creation.
- Design a multi-turn interactive, auto-planning, multi-agent dynamic collaboration system to generate high-quality, professional deliverables in both images and videos.
- Implement intent recognition, hierarchical planning, multimodal understanding and automatic self-reflection, multimodal deep search, RAG, and fine-grained selection of similar tools.
- Independently lead the automatic evaluation system development and reward functions design, focusing on confident evaluation and reward assingment for long-trajectory planning, tool invocation, and non-varifiable visual output generation.
- Take charge of the multimodal deep search module and domain-specific RAG modules implementation.

Circle Cat Research Department | Research Intern USA 2025.2 - China 2025.12

LLM for Muti-level Coding Error Classification
Role: Project Leader, First Author *Supervisor: Kazuma Hashimoto, Lingyu Gao*

- **Under review:** PyMETA: A Benchmark Dataset for Hierarchical Student Code Error Classification with Python-Interpreter-Based Labels
- Design a multi-level, fine-grained taxonomy for web-environment Python code error types in online compilers with LLM integration, and develop a benchmark based on 40,000 real-world Python code errors with multi-level classification, combining single-error and multi-error detection cases.
- Enhance LLM's multi-level error detection and classification through fine-tuning and prompting on multiple models, leveraging real error patterns and unstructured debugging logs from online compilers.
- Analyze and compare LLM performance on hierarchical error granularity and single-to-multi error detection tasks, examining biases, capability limits, and confidence across seven model architectures.

Multi-modal Time Series Conditional Generation from Unstructured Text***Supervisor: Kan Ren***

- **Accepted by ICML2025:** VerbalTS: Generating Time Series from Texts
- Innovatively design a framework to enable finer-grained cross-modal semantic alignment and control in generating time series data from unstructured text.
- Implement a Multi-view Noise Estimator that performs multi-resolution modeling and estimation of noise across temporal, spatial, and diffusion perspectives.
- Propose a Multi-focal Text Processor that decomposes text embeddings into multi-semantic levels (temporal, spatial, and diffusion stages), integrates learnable anchor vectors, and achieves fine-grained text alignment through reprogramming;
- Build an adapter module that gradually integrates and aligns multi-semantic textual information with the time series generation process for precise and hierarchical cross-modal alignment.
- Propose CTTP model and CTTP scores; Improve FID, J-FTSD, CTTP Score by 20%+ over baselines.

Unified Model for Multi-modal TS Generation, Prediction and Editing***Supervisor: Kan Ren***

- **Under review:** Unified Generative Modeling for Multimodal Time Series Analysis
- Propose GENTS: unifying time series conditional generation, forecasting, and editing into a masked-conditional generation problem, modeled within a unified framework.
- Design the unified noise estimator enabling cross-task generative capabilities and adaptive multimodal time series modeling.
- Introduce a domain-agnostic automated time series captioning method based on the statistical and morphological features of time series data, effectively addressing the scarcity of domain-specific multimodal paired data.

Publications & Manuscripts

[1] AVA-Encoder: Towards Agent-Native Video Representation Learning. **Under Review**

[Chuyue Li, Jinpeng Yu, Haozhe Wang, Tian Xueyun, Zhijing Zhang, Bingnan Li, Shuqi Gu, Kan Ren, Jiaming Liu, Ruihua Huang]

[2] VisionCreator: A Native Visual-Generation Agentic Model with Understanding, Thinking, Planning and Creation. **Accepted by CVPR2026 Findings.**

[Jinxiang Lai, Zexin Lu, Jiajun He, Rongwei Quan, Wenzhe Zhao, Qinyu Yang, qi chen, Qin Lin, **Chuyue Li**, Tao Gao, Song Guo, Qinglin Lu]

[3] What to Edit Next: Visually Aligned Image-Editing Follow-Up Suggestions in Conversational Systems

[Zhijing Zhang, Jinpeng Yu, Xin Song, Bingnan Li, **Chuyue Li**, Changhui Du, Xiaolin Fang, Jiaming Liu, Ruihua Huang]

[4] VerbalTS: Generating Time Series from Texts. **Accepted by ICML2025**

[Shuqi Gu, **Chuyue Li**, Baoyu Jing, Kan Ren]

[5] PyMETA: A Benchmark Dataset for Hierarchical Student Code Error Classification with Python-Interpreter-Based Labels. **Under review.**

[Chuyue Li, Ziqi Tang, Jingyi Wang, Yu Wu, Kazuma Hashimoto, Lingyu Gao]

[6] Unified Generative Modeling for Multimodal Time Series Analysis. **Under review.**

[Shuqi Gu, **Chuyue Li**, Yi Huang, Baoyu Jing, Yongxiang Zhao, Kan Ren]

Professional Skills

- **Languages:** Python, C/C++, Java, JavaScript, HTML, Bash, MATLAB, RISC-V
- **Tools:** PyTorch, Verl, ms-swift, vllm, Langgraph, Langchain, tsfresh, Gradio, HuggingFace, Docker, Kubernetes, Spring Boot, Coder, Bazel, Maven, Jenkins, MySQL, MongoDB, Gerrit, Linux, JUnit, Git, jQuery, GCP, Google Registry, Gitlab, Shell
- **English:** TOEFL 110 (R:27 L:27 S:28 W:28)